

Homework 4

Due: Sunday, 30th June 2024 at 1:00 PM

Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question (except the extra credit question) is worth 10 points.

Question 1.

A permutation of a set is an arrangement of its members into a sequence. If the set is $\{1, 2, 3\}$, then the possible permutations are: $(1, 2, 3)$, $(1, 3, 2)$, $(2, 1, 3)$, $(2, 3, 1)$, $(3, 1, 2)$, and $(3, 2, 1)$.

Hence the number of permutations of this set is 6.

- a) Find a recurrence relation for the number of permutations of a set with n elements.
- b) Use this recurrence relation to find the number of permutations of a set with n elements using iteration.

Question 2.

Find a recurrence relation to compute the number of bit strings. Write down a procedure that takes n as an input parameter and returns the number of bit strings. You can provide the pseudocode or implement it in your favorite programming language.

Question 3.

How can you modify the procedure so that instead of returning the number of bit strings, it returns a list of all bit strings? Please provide the procedure/program.

Question 4.

How can you verify given a bit string whether it contains two consecutive zeros? Please provide a procedure/program.

Question 5.

- a) Find a recurrence relation for the number of bit strings of length n that contain a pair of consecutive 0s.
- b) What are the initial conditions?
- c) How many bit strings of length seven contain two consecutive 0s?

Question 6.

- a) Find a recurrence relation for the number of bit strings of length n that contain three consecutive 0s.
- b) What are the initial conditions?
- c) How many bit strings of length seven contain three consecutive 0s?

For Questions 7-11, solve the recurrence relation together with the initial conditions given.

Question 7.

Solve $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$.

Question 8.

Solve $a_n = 4a_{n-2}$ for $n \geq 2$, $a_0 = 0$, $a_1 = 4$.

Question 9.

Solve $a_n = a_{n-2} / 4$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$.

Question 10.

Solve $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2$, $a_0 = 3$, $a_1 = 6$.

Question 11.

Solve $a_n = 7a_{n-1} - 10a_{n-2}$ for $n \geq 2$, $a_0 = 2$, $a_1 = 1$.

Question 12.

For integer multiplication, we solved the n -bit problem by using three solutions of $n/2$ -bit problems and additional cn operations. However, the analysis was very complicated. If we use a simpler analysis and solve the n -bit problem by using four solutions of $n/2$ -bit problems and additional c_1n operations, then what will be the recurrence? What will be the solution to that recurrence achieved from the simpler analysis?

Question 13.

Suppose that $f(n) = f(n/3) + 1$ when n is a positive integer divisible by 3, and $f(1) = 1$. Find
a) $f(3)$. b) $f(27)$. c) $f(729)$.

Question 14.

Find $f(n)$ when $n = 2^k$, where f satisfies the recurrence relation $f(n) = f(n/2) + 1$ with $f(1) = 1$.

Question 15.

Find $f(n)$ when $n = 3^k$, where f satisfies the recurrence relation $f(n) = 2f(n/3) + 4$ with $f(1) = 1$.

Extra credit.

Show that for any positive integer n , $(1-1/n)^n \leq 1/e$ where e is Euler's number and the base of natural logarithm.

Notice that, if you answer the extra credit question correctly, the grade of the extra credit will not show up on Gradescope. But we will add that grade in D2L at the end of the semester.